

Divyansh Shukla

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EDUCATION

Vellore Institute of Technology

B.Tech, Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Expected 2027

CGPA: 9.03 / 10

EXPERIENCE

Power Pulse India

2026 – Present

Machine Learning Intern

India

- Trained a deployment-aware YOLOv8n thermal detector on FLIR ADAS v2 (6 road-safety classes) to mAP50-95 0.420 and recall 0.604 from a 0.360 / 0.540 baseline, in 100 epochs against 150 after validation curves went flat past epoch 90; accepted 3 points of precision for 26% fewer missed pedestrians, the correct trade for driver assistance.
- Root-caused the accuracy ceiling to annotations rather than model capacity: crop audits showed labelled “trucks” were delivery vans also annotated *car* and *bus* (37% of true trucks drew no prediction, 24% car, 15% bus), compounded by unlearnable 2–6 px boxes and detail lost at 640px.
- Attributed the gain to small objects (82% of pedestrian labels under 32×32 px; recall 0.689 → 0.776) and costed it at 2.25× compute per frame (3.4 → 7.7 GMACs); tested a cheaper P2 detection head at native resolution, which recovered less of the gain and confirmed input resolution as the effective lever.

CERTIFICATIONS

NPTEL: Applied Accelerated AI, Deep Learning with Computer Vision | **Coursera:** Deep Learning Specialization | **AWS:** AWS Cloud Architecting

TECHNICAL SKILLS

Languages: Python, SQL, C++, CUDA

ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers, TRL, Unsloth, Ultralytics (YOLO), Stable-Baselines3, vLLM, llama.cpp

Deep Learning: diffusion models, flow matching, transformers, attention mechanisms (RoPE, AdaLN-Zero), classifier-free guidance, object detection, reinforcement learning (PPO), vision-language models

LLMs & Agents: LangChain, LangGraph, CrewAI, Google Agent Development Kit (ADK), Google Gen AI SDK, prompt engineering, structured output extraction, function calling, agentic workflows (planning, memory, tool use), LLM evaluation

Retrieval-Augmented Generation (RAG): EmbeddingGemma embeddings, pgvector, ChromaDB, similarity search

Cloud & Data: AWS (S3, EC2, Lambda, SageMaker, Redshift, EMR, ECS/ECR, IAM, CloudWatch), Google Cloud (BigQuery, Dataproc, Dataflow, Pub/Sub, Cloud Storage, Vertex AI), Apache Spark (managed on EMR and Dataproc), Modal, exploratory data analysis, data cleaning and visualization

AI Developer Tools: Claude Code, Gemini Antigravity, Gemini API

Simulation & Robotics: MuJoCo, Gymnasium, OpenCV, Raspberry Pi, embedded/edge inference

Backend & Infrastructure: FastAPI, Django, Flask, Weights & Biases, Hydra, Git, Linux, NumPy, pandas

PROJECTS

RoPE-DiT: Diffusion Transformer with 2D RoPE

github.com/divyanshuklai/Custom2DRoPEDiT

- Wrote a ~5M-parameter Diffusion Transformer from scratch in PyTorch: patchifier, timestep and class-label embedders, AdaLN-Zero conditioning, and a custom multi-head attention module.
- Built a 2D rotary positional embedding that rotates queries and keys by each patch’s height and width coordinates, splitting the head dimension into row and column halves so attention sees relative 2D distance.
- Trained on CIFAR-10 with a rectified flow objective and sampled with a hand-written Euler solver and classifier-free guidance.

Zero-Shot ICE RMA: Quadruped Locomotion RL

github.com/divyanshuklai/zero-shot-ice-rma

- Built a MuJoCo environment for the Unitree Go1 quadruped with 30-dim proprioceptive observations, 12-dim joint-position actions, fractal terrain, and the Rapid Motor Adaptation bioenergetics reward with curriculum-scheduled penalties.
- Debugged a PPO baseline that ran 42M steps without learning a gait, tracing the exploding value loss to unnormalized observations and penalty dominance.

Fashion Search: Conversational Product Discovery

github.com/divyanshuklai/fashion_search

- Built a FastAPI chat app answering natural-language product queries across 10 Indian e-commerce stores with Tavily search, Gemini 2.5 Flash extraction, and BeautifulSoup scraping.
- Wrote a priority-based image and price extractor that tries JSON-LD product schema, then Open Graph tags, then keyword-matched image tags, as the baseline for an agentic version with planning, memory, and tool use.

ACHIEVEMENTS

Won the internal (intra-campus) round of the Smart India Hackathon at Vellore Institute of Technology, 2024.